



Hawea Water Usage Processing

Hawea Groundwater modelling project

Report 1052-08-01 R1

Mike Kittridge (sub-contractor)
Lincoln Agritech Ltd

28 July 2022

LINCOLN AGRITECH LOCATIONS

Lincoln Agritech is a 100% subsidiary of Lincoln University and is based on campus at Lincoln University, near Christchurch, New Zealand. Our North Island office is located on the Ruakura Research Campus in Hamilton, New Zealand.

Lincoln/Head Office

Lincoln Agritech Limited / Engineering Drive / Lincoln University
PO Box 69133 / Lincoln / Christchurch 7640 / New Zealand

P +64 3 325 3700 / **F** +64 3 325 3725

Hamilton/Regional Office

Ruakura Science Centre / East Street
Hamilton 3240 / New Zealand

P +64 7 858 4840

CONFLICTS OF INTEREST, CONFIDENTIALITY & COPYRIGHT

Conflicts of Interest

We are not aware of any circumstance where a conflict of interest could arise. However, should a potential conflict of interest become evident, we would immediately bring this to the attention of our client and discuss appropriate action and mutually acceptable ways to move forward.

Confidentiality & Copyright

Lincoln Agritech reserves copyright in the concepts, statements and content of this project proposal. These may not be disclosed to a third party or used for any purpose other than the negotiation of a contract between Lincoln Agritech and its client.

DOCUMENT ACCEPTANCE

ACTION	NAME	SIGNED	DATE
Group Manager	Blair Miller		28/7/22

1. PURPOSE

The primary purpose of this work is to process the Otago Regional Council's (ORC) water use data for use as input for the Hawea groundwater modelling project. Secondly, providing a water use dataset that identifies data issues across the region would support regional scientific modelling and planning in a variety of ways.

The time period considered only includes data starting in 2014-07-01 and ending on 2021-06-30. This was due to the water consents at ORC not retaining adequate change history and the time period requirements of the groundwater modelling. Consequently, only current/active water consents were utilised.

This work was not scoped to "correct" existing data and the gap filling method is specifically designed for the purpose of hydrologic modelling rather than to perfectly recreate a water use time series that can be used for consent compliance.

2. GENERAL APPROACH

I took a three step approach to the data cleansing process:

1. Run the data through automatic checks to flag/filter data with issues and identify missing data.
2. Fill the gaps that already exist or were filtered out in step 1.
3. Pass the flagged data issues to compliance officers.

Parts 1 and 2 can be performed on the entire Otago Regional Council (ORC) dataset as they will be automated procedures. At the time of writing, I did not receive all of the water use data for Otago, so only the Hawea groundwater model domain was processed. Part 3 will be passed to compliance for a relatively quick manual check of the flagged water use data and the associated consent data with priority will be given to the larger consents.

3. METHODOLOGY

The automated checks and gap filling were based on existing work that was performed at Environment Canterbury (ECan) for their water use and allocation data. The open-source python package used for this is called [nz-allo-usage-tools](#). In addition to water usage data, consented allocation data and consent use type are required to perform the checks and fill gaps.

A key assumption that has been made was to distribute the allocation across multiple water meters. When there are multiple water meters per consent, the total allocation has been split evenly across all WAPs. For example, if there are three water meters on one consent with a total allocation rate of 90 l/s, then each water meter would be assigned an allocation of 30 l/s.

Another issue is that ORC does not explicitly assign an effective end date to expired consents, which means that when a consent is changed from “Current” to “Expired” there is no clear way to know when the consent ceased to be active. There may be indirect ways to determine parent/child relationships between consents, but those will only help in very limited circumstances. Consequently, I have only used Current/Active consents in this assessment. Using only the Current/Active consents means that the further we go back in time the more we underestimate the allocation and usage.

3.1 Automated checks

My experience working with the water usage data at ECan is that many of the standard statistics for outlier detection do not work adequately due to how highly skewed and irregular these datasets are.

Simple checks were performed for negative values and values that exceed abstraction that is physically possible. The next kind of check compared the consented allocation to the usage at multiple temporal scales. This looked at the usage/allocation ratios at the daily, monthly, and yearly scales. The defaults are 2, 3, and 2 respectively. These were found in Canterbury to be generous enough to retain real-looking usage values and exclude erroneous ones, but these values may need to be modified based on ORC water use data. Even with appropriate ratio values, it is still unavoidable to get false positives and negatives using these filters.

3.2 Gap filling

Next, abstraction/allocation ratios were calculated and lumped by month of the year and use type. These ratios were then applied to the water meters that did not have usage data to estimate the usage data by month. The algorithm prioritises assigning missing values from nearby abstractors from similar use types. The monthly estimates were resampled to daily using the pchip interpolation method in Python. This method was used to make the result more smooth and consequently more natural looking.

3.3 Pass flagged data to compliance officers

The data that has been flagged and categorised will be passed to compliance officers. Based on the type of issue, the compliance officer would need to either attempt to have the water use inspected or enquire with the consent holder about the data.

4. RESULTS

As there can be multiple water meters per consent, the rest of the report will present statistics based on these combined consent/water meter numbers. When the report uses the term “consent” or “consents”, it is implied that this means the consent/water meter combinations.

There were 77 consumptive water take consents starting in 2014-07-01 and 119 consents ending in 2021-06-30 within the Hawea groundwater model domain (Fig 1, Fig 2). Of those 119, 97 had water usage data. Of the consents with water usage data, they covered on average about 66% of the 7 year assessment period. The ratio of total consents to consents with available data has been relatively consistent over the years with a slow decline starting in 2020-07-01 until recent times.

Over the 7 year period, the consents with metered data accounts for about 76% of the total allocation volume (Fig 3).

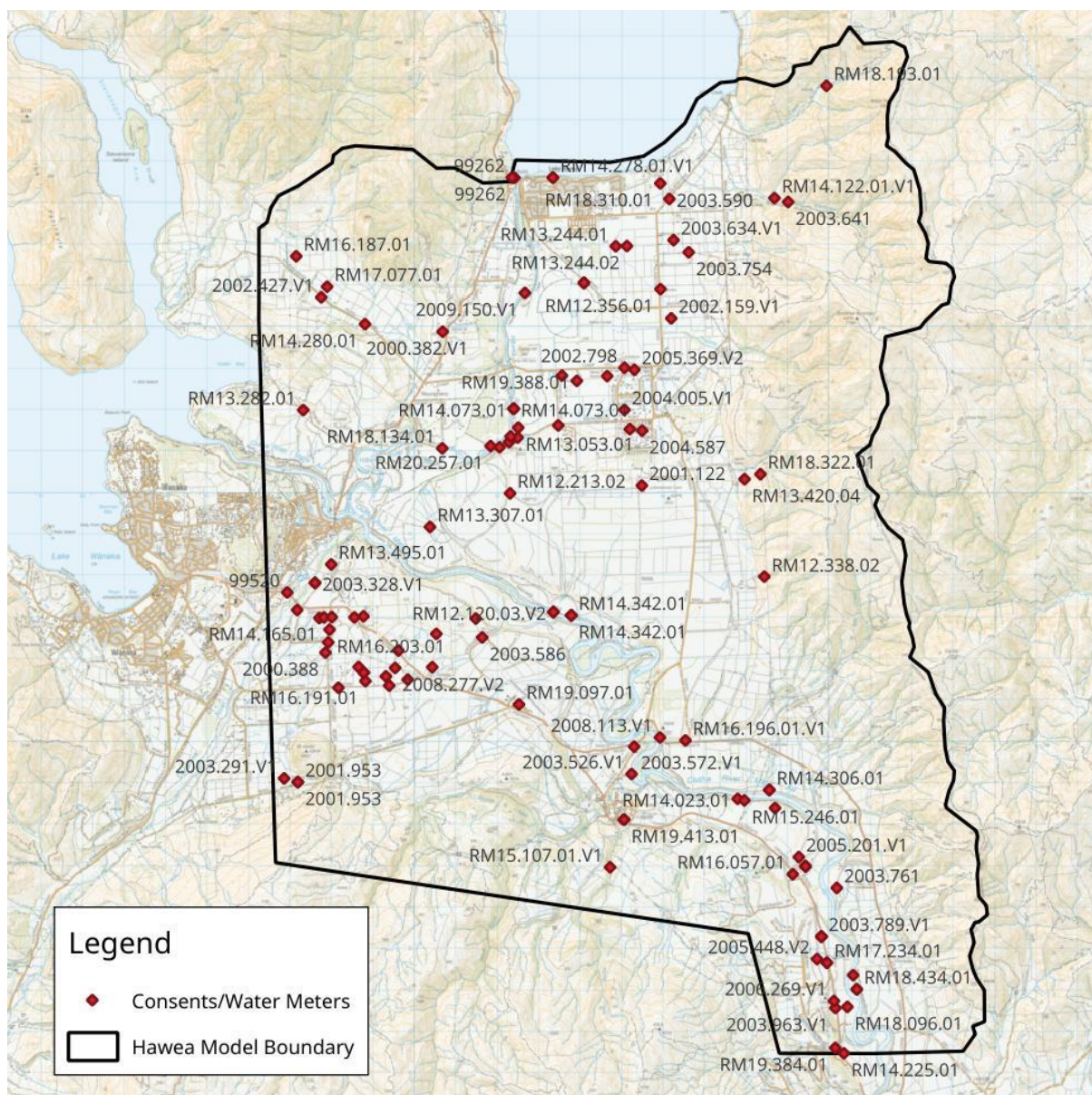


Figure 1. Model domain with the consent/water meter locations as of 2021-06-30.

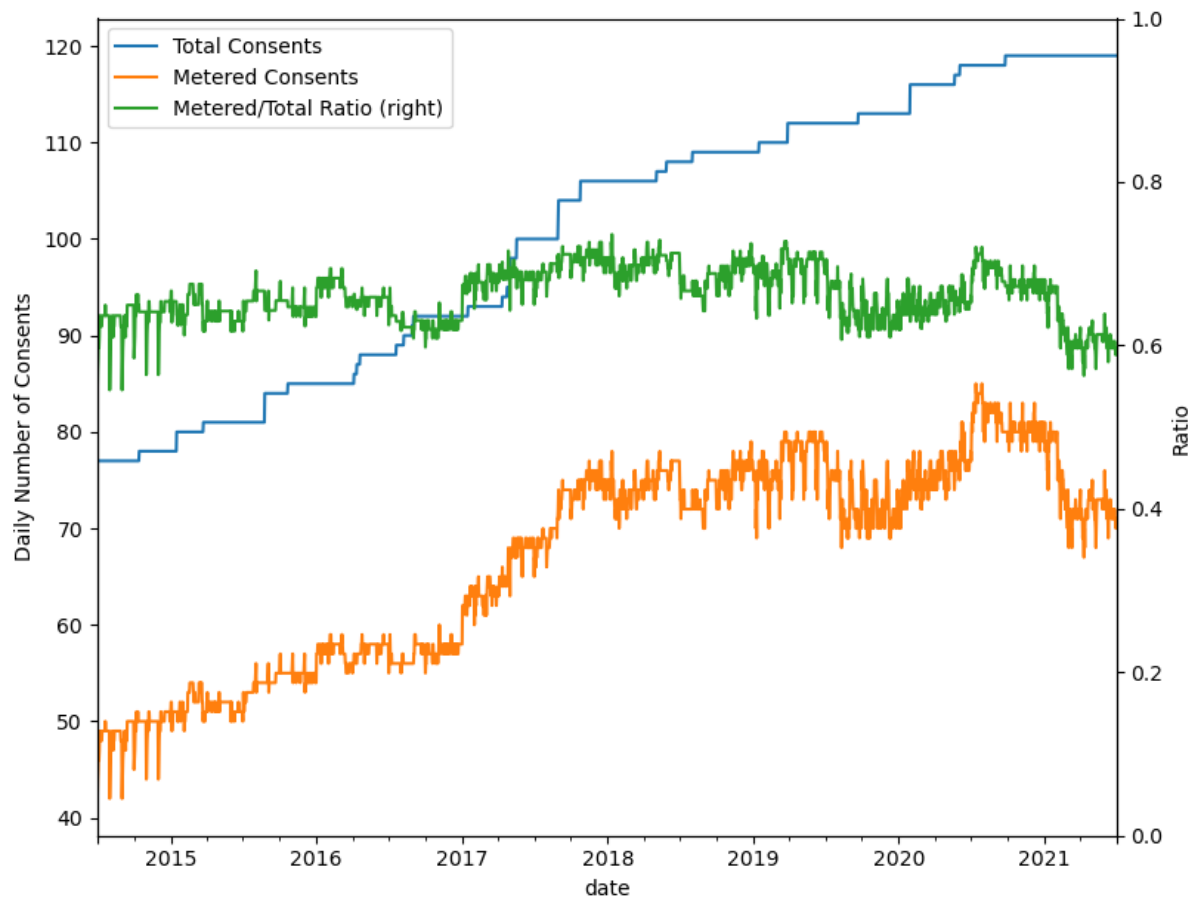


Figure 2. The daily number of consents (total and metered with water usage data) and the associated ratio from 2014-2021 within the Hawea model domain.

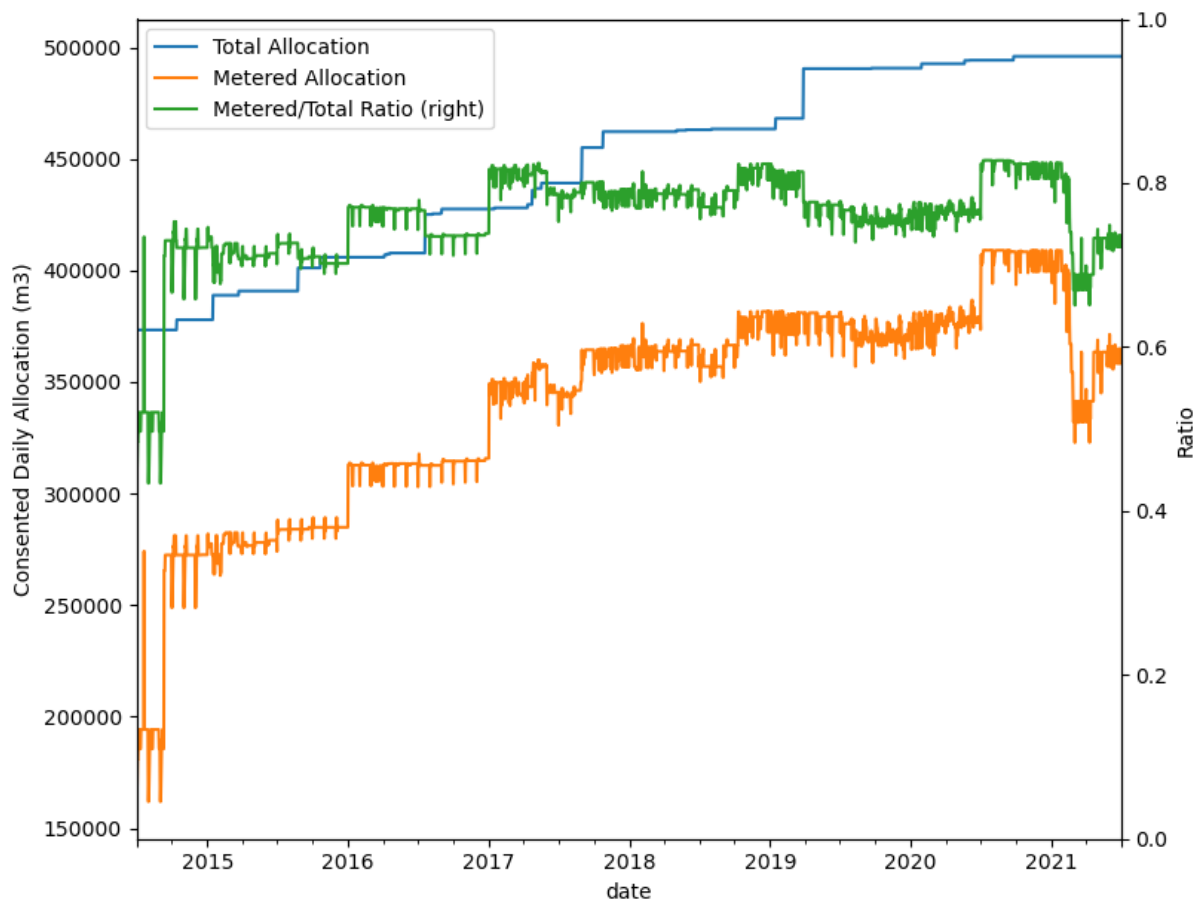


Figure 3. The daily consented allocation (total and metered with water usage data) and the associated ratio from 2014-2021 within the Hawea model domain.

4.1 Checks

There were no consents that had negative values. On a daily scale, there were 19 consents covering 0.9% of the record that had values over two times their consented volumes. On a monthly scale, there were 9 consents covering 1.1% of the record that had values over three times their consented volumes. On a yearly scale (running from July to June), there were 11 consents covering 3.0% of the record that had values over two times their consented volumes.

4.2 Gap Filling

The gap filling for the groundwater water modelling was performed at the daily scale. The gaps that were filled were from both the missing data and the data found in the Checks section to be above the usage/consent threshold (0.9% of the record). The gap filling procedure added 21% more usage to the existing data.

5. DESCRIPTION OF THE OUTPUTS

Seven outputs as csv files were provided. These include water_permit_meter_daily_data_2022-07-20.csv, water_permit_meter_daily_exceedences_2022-07-20.csv, water_permit_meter_exceedence_summary_2022-07-20.csv, water_permit_meter_monthly_data_2022-07-20.csv, water_permit_meter_monthly_exceedences_2022-07-20.csv, water_permit_meter_yearly_data_2022-07-20.csv, and water_permit_meter_yearly_exceedences_2022-07-20.csv.

The following table describes the fields of the files.

file name	field	description
water_permit_meter_exceedence_summary_2022-07-20.csv	permit_id	The water permit id
water_permit_meter_exceedence_summary_2022-07-20.csv	water_meter_no	The water meter number
water_permit_meter_exceedence_summary_2022-07-20.csv	percent missing days	The percent missing days for the entire record
water_permit_meter_exceedence_summary_2022-07-20.csv	percent days exceedence	The percent days exceeding 2 times the allocation for the entire record
water_permit_meter_exceedence_summary_2022-07-20.csv	percent months exceedence	The percent months exceeding 3 times the allocation for the entire record
water_permit_meter_exceedence_summary_2022-07-20.csv	percent years exceedence	The percent years exceeding 3 times the allocation for the entire record
water_permit_meter_daily_exceedences_2022-07-20.csv	permit_id	The water permit id
water_permit_meter_daily_exceedences_2022-07-20.csv	water_meter_no	The water meter number
water_permit_meter_daily_exceedences_2022-07-20.csv	date	The date of the value

_2022-07-20.csv		
water_permit_meter_daily_exceedences_2022-07-20.csv	total_allo	The total allocation in m3 of the permit, water meter, and date
water_permit_meter_daily_exceedences_2022-07-20.csv	total_usage	The total usage in m3 of the permit, water meter, and date
water_permit_meter_monthly_exceedences_2022-07-20.csv	permit_id	The water permit id
water_permit_meter_monthly_exceedences_2022-07-20.csv	water_meter_no	The water meter number
water_permit_meter_monthly_exceedences_2022-07-20.csv	date	The date of the value
water_permit_meter_monthly_exceedences_2022-07-20.csv	total_allo	The total allocation in m3 of the permit, water meter, and date
water_permit_meter_monthly_exceedences_2022-07-20.csv	total_usage	The total usage in m3 of the permit, water meter, and date
water_permit_meter_yearly_exceedences_2022-07-20.csv	permit_id	The water permit id
water_permit_meter_yearly_exceedences_2022-07-20.csv	water_meter_no	The water meter number
water_permit_meter_yearly_exceedences_2022-07-20.csv	date	The date of the value
water_permit_meter_yearly_exceedences_2022-07-20.csv	total_allo	The total allocation in m3 of the permit, water meter, and date
water_permit_meter_yearly_exceedences_2022-07-20.csv	total_usage	The total usage in m3 of the permit, water meter, and date
water_permit_meter_daily_data_2022-07-20.csv	permit_id	The water permit id

water_permit_meter_daily_data_2022-07-20.csv	water_meter_no	The water meter number
water_permit_meter_daily_data_2022-07-20.csv	date	The date of the value
water_permit_meter_daily_data_2022-07-20.csv	total_allo	The total allocation in m3 of the permit, water meter, and date
water_permit_meter_daily_data_2022-07-20.csv	sw_allo	The surface water allocation in m3 of the permit, water meter, and date
water_permit_meter_daily_data_2022-07-20.csv	gw_allo	The groundwater allocation in m3 of the permit, water meter, and date
water_permit_meter_daily_data_2022-07-20.csv	total_metered_allo	The total metered allocation in m3 of the permit, water meter, and date. Metered allocation means the amount of allocation that has water usage data associated with it.
water_permit_meter_daily_data_2022-07-20.csv	sw_metered_allo	The surface water metered allocation in m3 of the permit, water meter, and date. Metered allocation means the amount of allocation that has water usage data associated with it.
water_permit_meter_daily_data_2022-07-20.csv	gw_metered_allo	The groundwater metered allocation in m3 of the permit, water meter, and date. Metered allocation means the amount of allocation that has water usage data associated with it.
water_permit_meter_daily_data_2022-07-20.csv	total_usage	The total usage in m3 of the permit, water meter, and date
water_permit_meter_daily_data_2022-07-20.csv	sw_allo_usage	The surface water usage in m3 of the permit, water meter, and date
water_permit_meter_daily_data_2022-07-20.csv	gw_allo_usage	The groundwater usage in m3 of the permit, water meter, and date
water_permit_meter_daily_data_2022-07-20.csv	total_usage_est	The total usage plus the estimated missing usage in m3 of the permit, water meter, and date
water_permit_meter_daily_data_2022-07-20.csv	sw_allo_usage_est	The surface water usage plus the estimated missing usage in m3 of the permit, water meter, and date
water_permit_meter_daily_data_2022-	gw_allo_usage_est	The groundwater usage plus the estimated missing usage in m3 of the permit, water

07-20.csv		meter, and date
water_permit_meter_monthly_data_2022-07-20.csv	permit_id	The water permit id
water_permit_meter_monthly_data_2022-07-20.csv	water_meter_no	The water meter number
water_permit_meter_monthly_data_2022-07-20.csv	date	The date of the value
water_permit_meter_monthly_data_2022-07-20.csv	total_allo	The total allocation in m3 of the permit, water meter, and date
water_permit_meter_monthly_data_2022-07-20.csv	sw_allo	The surface water allocation in m3 of the permit, water meter, and date
water_permit_meter_monthly_data_2022-07-20.csv	gw_allo	The groundwater allocation in m3 of the permit, water meter, and date
water_permit_meter_monthly_data_2022-07-20.csv	total_metered_allo	The total metered allocation in m3 of the permit, water meter, and date. Metered allocation means the amount of allocation that has water usage data associated with it.
water_permit_meter_monthly_data_2022-07-20.csv	sw_metered_allo	The surface water metered allocation in m3 of the permit, water meter, and date. Metered allocation means the amount of allocation that has water usage data associated with it.
water_permit_meter_monthly_data_2022-07-20.csv	gw_metered_allo	The groundwater metered allocation in m3 of the permit, water meter, and date. Metered allocation means the amount of allocation that has water usage data associated with it.
water_permit_meter_monthly_data_2022-07-20.csv	total_usage	The total usage in m3 of the permit, water meter, and date
water_permit_meter_monthly_data_2022-07-20.csv	sw_allo_usage	The surface water usage in m3 of the permit, water meter, and date
water_permit_meter_monthly_data_2022-07-20.csv	gw_allo_usage	The groundwater usage in m3 of the permit, water meter, and date
water_permit_meter_yearly_data_2022-07-20.csv	permit_id	The water permit id

water_permit_meter_yearly_data_2022-07-20.csv	water_meter_no	The water meter number
water_permit_meter_yearly_data_2022-07-20.csv	date	The date of the value
water_permit_meter_yearly_data_2022-07-20.csv	total_allo	The total allocation in m3 of the permit, water meter, and date
water_permit_meter_yearly_data_2022-07-20.csv	sw_allo	The surface water allocation in m3 of the permit, water meter, and date
water_permit_meter_yearly_data_2022-07-20.csv	gw_allo	The groundwater allocation in m3 of the permit, water meter, and date
water_permit_meter_yearly_data_2022-07-20.csv	total_metered_allo	The total metered allocation in m3 of the permit, water meter, and date. Metered allocation means the amount of allocation that has water usage data associated with it.
water_permit_meter_yearly_data_2022-07-20.csv	sw_metered_allo	The surface water metered allocation in m3 of the permit, water meter, and date. Metered allocation means the amount of allocation that has water usage data associated with it.
water_permit_meter_yearly_data_2022-07-20.csv	gw_metered_allo	The groundwater metered allocation in m3 of the permit, water meter, and date. Metered allocation means the amount of allocation that has water usage data associated with it.
water_permit_meter_yearly_data_2022-07-20.csv	total_usage	The total usage in m3 of the permit, water meter, and date
water_permit_meter_yearly_data_2022-07-20.csv	sw_allo_usage	The surface water usage in m3 of the permit, water meter, and date
water_permit_meter_yearly_data_2022-07-20.csv	gw_allo_usage	The groundwater usage in m3 of the permit, water meter, and date

